

1995 HKCEE MATHS Paper II

Label 1 Round off the number 0.044449 to 3 significant figures.

- A. 0.04
B. 0.044
C. 0.045
D. 0.0444
E. 0.0445

Label 2 If $\frac{x+y}{xy} = 1$, then $y =$

- A. $\frac{1-x}{x}$
B. $\frac{x-1}{x}$
C. $\frac{x}{1-x}$
D. $\frac{x}{x-1}$
E. $\frac{1-x}{1+x}$

Label 3 If $f(x) = x^{99} + 99x + k$ is divisible by $x + 1$, then $k =$

- A. -100
B. -98
C. 98
D. 100
E. 198

Label 4 Simplify $\left(\frac{a^6}{b^{12}}\right)^{-\frac{2}{3}}$

- A. $\frac{b^8}{a^4}$
B. $\frac{b^{18}}{a^9}$
C. $\frac{a^4}{b^8}$
D. $\frac{a^9}{b^{18}}$
E. $\frac{1}{a^4 b^{12}}$

Label 5 $\frac{1}{2+\sqrt{6}} - \frac{1}{2-\sqrt{6}} =$

- A. $-\sqrt{6}$
B. $-\frac{\sqrt{6}}{2}$
C. 0
D. $\frac{\sqrt{6}}{2}$
E. $\sqrt{6}$

Label 6 The L.C.M. of $x^3 - x$ and $x^4 - 1$ is

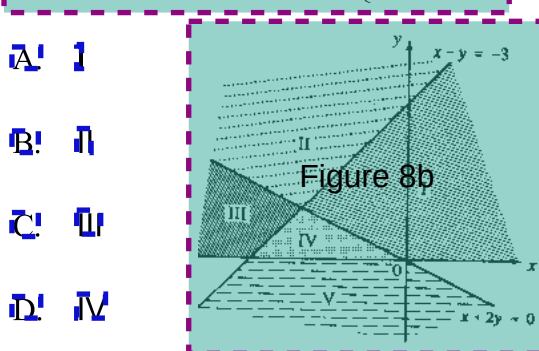
- A. $x-1$
B. $(x-1)(x+1)$
C. $x(x-1)(x+1)(x^2+1)$
D. $(x-1)(x+1)(x^2+1)(x^2+x+1)$
E. $x(x-1)^2(x+1)^2(x^2+1)$

Label 7 Solve the simultaneous equations:

- A. $x = -\frac{1}{2}, y = -12$
B. $x = -\frac{1}{2}, y = 12$
C. $x = \frac{1}{2}, y = -12$
D. $x = \frac{1}{2}, y = 12$
E. $x = \frac{5}{24}, y = -\frac{7}{2}$

Label 8 Which of the following shaded regions

represents the solution of



- A. I
B. II
C. III
D. IV
E. V